Reliable and Invalid: Anatomy of a Gamed Training Task

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Abstract

Verifiable rewards are trusted because they are deterministic: a program either awards the points or it does not. We post-train Qwen2.5-7B-Instruct with GRPO on NYT Connections under a reward with a cheap structural component (well-formed output) and an expensive semantic component (correct groupings). On a strictly chronological test set of 162 puzzles (648 group slots), scored in one serving session, capability runs $26 \to 52 \to 80 \to 4$ slots across the untrained Instruct model, supervised fine-tuning, GRPO step 50, and the GRPO endpoint: paired stage gains of $+0.161$ and $+0.173$ on the 0–4 scale, then a fall to within $3\times$ of the 1.4-slot chance floor of a uniformly drawn valid partition, below the untrained model, reproduced across three seeds. The *training* reward reads the destruction as mastery: it reaches its 1.6 ceiling by step 125 with zero across-rollout variance and stays there. Scoring the endpoint offline on its own training puzzles — 21 of 3,228 slots — rules out memorization and exposes the cause: a one-line bug in our GRPO data loader presented each board's sixteen words in answer-key order, so the training task, as presented, was solvable by copying the words four at a time. The collapsed policy does exactly that — 91.4% of the groups it emits on training boards, and 92.4% on test boards, are consecutive quadruples of its prompt; a copy of a shuffled board scores at or below the chance floor, which is why the endpoint's score collapses, and its small margin above that floor comes from the residual non-copy responses. Entropy collapse, cross-seed convergence, and the reward ceiling are signatures of one deterministic copy rule. The reward function was correct on every string it scored; the task presentation was gameable, and every in-sample signal read the gaming as mastery. The same reward held out, and three blind LLM judges, catch the reversal; four lines of string matching identify the rule; and rerunning the recipe with the one-line fix, the same reward trains the best policies in this study — 148, 135 and 129 of 648 slots across three seeds, each about five times the untrained model, with no collapse, no copying, and every in-sample signal reading honestly. Supervised fine-tuning on the leaked prompts alone converges to a perfect copier: the gameable object was the task, and it gamed every learner we gave it. Determinism certifies the scorer, not the construct: hold the reward out.

1 Introduction

Post-training against verifiable rewards is attractive for a reason that sounds decisive: the reward is a program. It returns exactly what it was written to return, and the argument that this buys immunity from reward hacking[1] rests on that reliability. This paper is about the gap reliability does not close: the gap between the reward and the construct it stands in for. We train on NYT Connections — sixteen words, partitioned into four groups of four — chosen because well-formedness and correctness are

[1]We follow the common usage *reward hacking* throughout; Skalse et al. [2022], whose formalization we adopt, publish it under the name *reward gaming*.

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unusually separable: a response can be a perfectly valid partition and completely wrong. Our reward pays for both; two of its five components can be checked by reading the string, three require the answer key.

For fifty steps it works. In one serving session, held-out groups recovered run $26 \to 52 \to 80$ across the untrained Instruct model, the supervised checkpoint GRPO is initialized from, and step 50 — paired stage gains of $+0.161$ $[+0.056, +0.272]$ and $+0.173$ $[+0.074, +0.284]$, the reinforcement-learning stage contributing as much as supervised fine-tuning (§7). Over the following 353 steps the same run falls to 4 slots, below the untrained model and to within $3\times$ of the 1.4-slot floor of a uniformly drawn valid partition, while the invalid-output rate falls by an order of magnitude and the training reward reaches its 1.6 maximum and averages 1.5814 thereafter, with zero variance across the $K = 8$ rollouts on $\sim$95% of prompts. A practitioner watching that curve saw a solved task. The curve was not lying about the reward; it was lying about the task. A control run scoring the endpoint on its own training puzzles returned 21 of 3,228 slots — not memorization — and led to the mechanism: our GRPO data loader, unlike our SFT and evaluation code, presented each board's words in answer-key order, and the converged policy simply copies the prompt four words at a time (91–92% of emitted groups; §7). The failure is not in the reward function, which is correct; it is in what the training distribution let the reward certify, and in the practice of reading the certificate in sample. The setting matters: in RL post-training the training distribution is regenerated by the trainer at every step, so a leak like ours lives in code rather than in a dataset — where none of the auditing reflexes the shortcut-learning literature built will look for it.

Contributions. A fully verifiable case study in which GRPO improves and then destroys the target construct, ending below the untrained model across three seeds (§3); a complete post-hoc diagnosis: an answer-ordered prompt bug made the training task gameable, and the converged policy is a prompt-order copier on 91–92% of its emitted groups (§7); an account in which entropy collapse, the reward ceiling, and cross-seed convergence are signatures of that one rule (§5), surviving a normalization ablation (§6); the lesson demonstrated end to end: every in-sample signal read gaming as mastery; the reward held out, and three blind judges, caught it (§7–§9); and the diagnosis closed by intervention: with the fix, the recipe trains to 148/135/129 across three seeds with no collapse, while SFT on the leaked prompts alone converges to a perfect copier — the attractor is the task's, not the algorithm's (§8).

2 Task, pipeline, and the reward's asymmetry

Task and data. 1,078 dated NYT Connections puzzles (1,077 after validation), split strictly chronologically: 807 train (2023-06-12 to 2025-08-27), 108 validation, 162 test (2025-12-15 to 2026-05-29); every test puzzle postdates every training puzzle. Date regressions of groups-correct across the test window are flat for both base and the trained policy (values in the repository): contamination controlled, though the intervals are wide and only large drift would show.

Pipeline. Qwen2.5-7B-Instruct (and 1.5B for the scale contrast), rank-16 LoRA ($\alpha = 32$, dropout 0.05, all linear modules) on a 4-bit base. SFT for 3 epochs on the training split (learning rate 10^{-4}, batch size 2, gradient accumulation 8), then GRPO [Shao et al., 2024] from the SFT checkpoint: $K = 8$ sampled completions per puzzle at temperature 0.9, group-relative advantages, $\beta = 0.001$ against the SFT reference, learning rate 5×10^{-6}, batch size 2 with gradient accumulation 8, one epoch (403 optimizer steps) at 7B.

The leak, disclosed up front. SFT data and every evaluation in this paper present each board's sixteen words deterministically shuffled (seeded by puzzle id), precisely so group order is not a giveaway. Our GRPO loader did not: a one-line bug built training prompts from the answer key, so during reinforcement learning — only there — the words arrived grouped in answer order: every GRPO prompt was solvable by a positional copy rule worth the full 1.6. Discovered post hoc (§7); disclosed here because nothing in the training-time sections can be read without it. Held-out measurements are unaffected: evaluation prompts were always shuffled. §8 reruns the recipe with the fix.

The reward. Table 1 is the crux of the paper.

Table 1: The reward decomposition. Two components are verifiable from the emitted string alone; three require the answer key. The policy is trained on the sum.

Component	Value	Cost to check
Parses as four groups of four	$+0.1$	**cheap** — read the string
Correct groups	$+1.0 \times (\text{correct}/4)$	needs the answer key
All four groups correct	$+0.5$	needs the answer key
Group with exactly 3 correct words	$+0.05$	needs the answer key
Invalid output	-0.1	**cheap** — read the string

Maximum 1.6. The model can raise its score by getting better at the puzzle or by getting better at emitting well-formed output; only one is the construct. That asymmetry — cheap and expensive components, independently satisfiable — is the design property this paper studies.

3 The arc on held-out data

Table 2 reports all four arms of the reported 7B run on the 162-puzzle test set, greedy, one vLLM session so every comparison is within-session; the supervised checkpoint is included because it, not the untrained model, is what GRPO initialized from.

Table 2: 7B on the test split (162 puzzles, 648 group slots; all four arms in one serving session; greedy). Means on the 0–4 scale are exact ratios of the count column. Bootstrap intervals are percentile over puzzles, $B = 1{,}000$ resamples, seed 0. This is the session §7 also scores on the training split.

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward	Solves
Instruct (untrained)	0.1605 $[0.105, 0.222]$	26	6.8%	0.165	0/162
SFT	0.3210 $[0.216, 0.432]$	52	22.2%	0.191	2/162
GRPO, step 50	**0.4938** $[0.383, 0.623]$	**80**	**16.0%**	**0.252**	2/162
GRPO, step 403	0.0247 $[0.000, 0.056]$	4	0.6%	0.125	0/162

Two reference points frame the table. A uniformly drawn valid partition recovers $4 \times 5775/2{,}627{,}625 = 0.0088$ groups per puzzle, or **1.4 slots** across this split; the step-403 endpoint sits at 2.8 times that floor — and §7 shows why: its copied responses score at or below chance, and the excess over the floor rides on the few responses that are not copies. Second, because all four arms sit in one session, the stage accounting is paired: $26 \to 52 \to 80 \to 4$, the supervised stage contributing $+0.1605$ $[+0.0556, +0.2716]$ and the reinforcement-learning stage $+0.1728$ $[+0.0741, +0.2840]$ — near-equal gains, both intervals excluding zero.

The method works, then reverses. Both movements are paired within Table 2's session: the climb to step 50 is the two stage gains just given, each excluding zero, and step 403 falls below the untrained model by -0.1358 $[-0.1975, -0.0802]$. An earlier three-arm session read the same two contrasts consistently (values in the repository); means in Table 2 are exact ratios of the count column. Two details locate the mechanism. Held-out mean reward reads 0.252 at step 50 and 0.125 at step 403 against base's 0.165 (the later-session step-100 point is in Figure 2), while the training reward reaches its ceiling by step 125 (§5): the same reward ranks the checkpoints correctly out of sample and fails to on the training curve — that contrast, not the reward's determinism, is this paper's subject. And the step-50 policy is worse than base on structure (16.0% invalid against 6.8%) but *not* worse than its actual initialization, the SFT arm at 22.2% — so what is specific to the phase after step 50 is the semantic collapse, not the structural climb, which was already more than half complete before the peak (§5).

How wide is the peak? In a separate session, step 100 evaluates at 76/648 to a step-50 re-serve's 80/648, paired -0.0247 $[-0.148, +0.093]$: uninformative rather than equivalence (the interval spans the whole step-50 effect; no margin was pre-specified). The ablation trajectory suggests decline by step 100; all both sessions agree on is that the collapse falls in 100–150 (§5).

Reproducibility across serving sessions. Base and the final checkpoint reproduce per-puzzle *exactly* across vLLM sessions (a deterministic copier and a converged base leave no argmax ties to resolve); step 50 does not (77–80 of 648), which is why every step-50 number carries its serving session, and SFT decodes 52–56 (Appendix C). No conclusion turns on the differences.

Robustness of the endpoint. Three independent 7B seeds land at 4, 7 and 11 recovered slots of 648 (0.045 ± 0.022 on the 0–4 scale) — all below base's 26. Their LoRA update directions agree pairwise (cosine $+0.67$ to $+0.69$) — three runs finding the same attractor (§7); seeds share an initialization and dataset, so no random-direction null applies. Best-of-16 sampling does not rescue the converged policy: a copier has nothing new to sample (single-seed, illustrative contrasts in Appendix C).

Conditioning on well-formed output widens the gap rather than explaining it. Is the effect an artifact of validity, since invalid generations score zero groups and validity is what GRPO improved? Restricting to structurally valid generations at 7B, each arm's conditional computed within a single serving session (the SFT arm's is its 56-slot session): base **0.172** ($n = 151$ of 162), SFT **0.444** ($n = 126$), GRPO **0.025** ($n = 161$). This is a rescaling and we label it as one (each conditional is the unconditional mean divided by the validity rate, so the gap widens automatically when the low-scoring arm is the more valid, as here); it licenses only the negative claim we need: the deficit survives with formatting held fixed. Among answers that parse, the converged policy recovers 4 group slots where the untrained model recovers 26.

The loss shows no stratum gradient we can resolve. Relative loss from the untrained model to the endpoint runs 67–100% across all four stratum labels (Appendix D) — registered pre-diagnosis as a failed prediction of selective inference; post-diagnosis, the copy rule's expected signature, since a positional rule is blind to stratum. With 21–69 puzzles per stratum and base rates as low as one slot, the design cannot separate uniform loss from a gradient: consistent with stratum-blindness, not evidence for it.

4 The scale contrast at 1.5B

At 1.5B the instrument reads nothing. The untrained model recovers one group slot of 648, GRPO one, and SFT — the best arm — two, against Section 3's 1.4-slot chance floor. No arm is distinguishable from that floor (the exact intervals of Appendix F contain it), so no claim about the construct is available at this scale. What is visible is the reward driving its verifiable component: invalid output $32.1\% \to 2.5\%$, mean reward $0.049 \to 0.113$ (Appendix F). We report it as an observation: epoch-asymmetric runs, no 1.5B trajectory — and "untouched at 1.5B" would be an endpoint claim, which this paper shows cannot settle.

5 Trajectory of the collapse: entropy, KL, and the training signal

Figure 1 scores checkpoints of the reported run on the $n = 100$ temperature-0.9 *entropy sweep* (denominators of 400; full per-checkpoint values in Appendix B); the ablation's *checkpoint sweep* (§6) is $n = 108$, greedy, /432 — different measurements, never mixed.

The semantic peak has a location; the structural climb does not. Semantic rises $0.0325 \to 0.0950$ from the SFT initialization to step 50 ($13/400 \to 38/400$ slots), falls to 0.0075 by step 150, and holds at 0.0100 from 200 on. Structural (valid-partition fraction, not shown) climbs $0.37 \to 0.68 \to 0.64 \to$ 0.95–0.96 across init/50/100/150 on — one reversal, at 100 — with 0.31 of its 0.59 total gain banked by step 50, concurrent with the semantic peak.

The trajectory is not a trade with a turning point: a broadly monotone structural climb (one reversal, at 100) with a semantic hump on top, and only the hump has a location. One gap qualifies the shape: no checkpoint exists between steps 100 and 150 — 81% of the KL displacement ($8.12 \to 46.42$ nats) — so the descent is interpolated, not measured.

Entropy collapse is the rule's signature. Policy entropy falls 13.4-fold between steps 50 and 150, the window in which the semantic score falls 12.7-fold; by step 150 the policy is effectively deterministic. With §7's diagnosis in hand the reading is direct: a positional copy rule is deterministic,

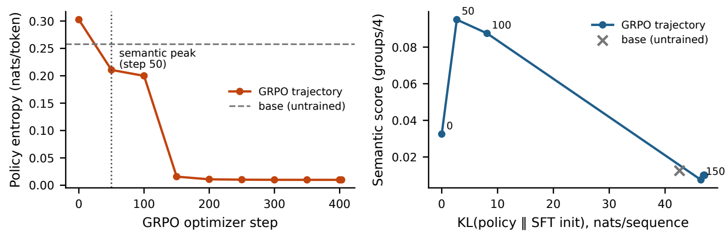

Figure 1: The mechanism, on the $n = 100$ temperature-0.9 entropy sweep (reported 7B run, seed 0). **Left:** policy entropy against optimizer step — a 30-fold collapse, with the cliff between steps 100 and 150. **Right:** semantic score against KL from the SFT initialization: the peak arrives by 2.70 of an eventual 47.05 nats (5.7%; step 50 is the earliest checkpoint retained), the remainder undoes it. The untrained model is an off-trajectory reference at 42.60 nats: KL measures displacement, not damage (§5).

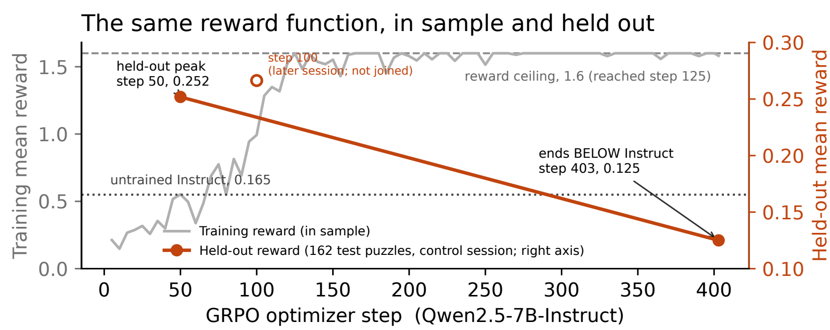

Figure 2: The same reward, in sample and held out (reported 7B run, seed 0). Training reward (gray; sampled T=0.9, K=8; TRL 5-step averages; measured on the answer-ordered prompts of §2) first touches its 1.6 ceiling at step 125, averaging 1.5814 thereafter. Held-out mean reward (162 test puzzles; markers; greedy; the control session of Table 3; right axis, note the scale) peaks at 0.252 at step 50 and ends at 0.125, below the untrained model's 0.165 — the dashed span between the anchors is unmeasured; the hollow step-100 point is a later session and is not joined.

blind to the board's content, and reliably well-formed: entropy collapse and semantic destruction read as two views of one convergence — a reading made after the diagnosis, and tested by intervention: remove the leak and entropy never collapses (§8) — which establishes the leak's necessity, not this finer reading; the leaked-SFT cell of §8 adds its sufficiency without RL. The structural climb is not — over half of it ($0.37 \rightarrow 0.68$ by step 50) predates measurable copying (0.3% at step 50 by the strict all-four-quadruples rule of Appendix D; a graded partial-adherence statistic could date copying earlier, which is why we write *measurable*): format gains the rule later locks in. What the collapse removes is within-prompt variation — nothing new to sample (the pass@16 result), no group-relative learning signal (the zero-advantage fraction). And what survives on held-out boards is measured twice over: the cheap structural term ($0.1(1 - 2 \cdot \text{invalid})$) is 52%/27%/79% of held-out reward across base/step 50/endpoint, and the copy rule is visible in 90.7% of the endpoint's test generations (§7).

KL is the budget view, and it is a path length, not a damage scale. The step-50 checkpoint sits 2.70 nats from the SFT initialization and is, jointly with step 100 (§3), the best policy this run produced on held-out data. The run spent 47.05: 80 slots at 5.7% of the displacement, 4 at all of it. Two qualifications keep this a path description, not a scale: step 50 is the earliest checkpoint retained, so "at 5.7%" means *at or before* it; and KL measures displacement, not damage — the untrained model sits 42.60 nats out on the same estimator while scoring 0.0125 on this sweep (Appendix B), so base is off this trajectory and 5.7% dates the peak within this run only. The sweep also compresses base's margin over the endpoint (5/400 vs. 4/400 here; 26 vs. 4 on the greedy test instrument): temperature-0.9 sampling costs a high-entropy policy far more than a collapsed one.

The training signal, read from the trainer's own logs. Figure 2 shows the paper's central contrast on one axis: the training reward first touches the reward function's ceiling of 1.6000 at step 125 and averages 1.5814 over the logged points from then on, with excursions that thin as training proceeds: the deepest is 1.4300 at step 155, and the count of logged points at exactly 1.6000 rises from 14 of 28 in the first half to 24 of 29 in the second. We describe rather than test — the series is autocorrelated, the metric bounded, no trend statistic we computed is identifiable, and part of the rise is the instrument (below). What the figure cannot show is TRL's own `frac_reward_zero_std` diagnostic: the fraction of prompts whose eight sampled rewards are identical — zero group-relative advantage. It rises from 0.0 at step 50 to 1.0 at step 125 and averages 0.946 over the remainder of training, so from step 125 onward the optimizer receives no learning signal on nearly every prompt. The saturation is legible under the diagnosis: on answer-ordered prompts (§2) a copy rule earns 1.6 on every board, seen or unseen — consistent with saturation beginning mid-epoch, on first-visit batches, and with the vanishing variance. Read as diagnostics, both in-sample signals bracket the collapse — the mean's climb to ceiling (0.9931 $\rightarrow$ 1.6000) and the variance's fall to zero both land between logged steps 100 and 125 — and neither signs it: saturation with zero spread is what mastery and degeneracy alike look like. Part of the climb is the instrument: as entropy collapses, temperature-0.9 sampling converges on greedy, so some of the rise is the sampling penalty vanishing (the sweep-vs-test gap, reversed). The remaining 278 steps of movement while $\sim$95% of prompts carry zero advantage are consistent with continued sharpening of the rule on the residual prompts that still do; the weight-space counterpart of that push we have not traced.

β was numerically inert, and the bracket is not an experiment. An earlier configuration at learning rate 10^{-6} and $\beta = 0.04$ froze the policy (KL ≈ 0.001; greedy outputs byte-identical to SFT); the reported one overshoots. The pair differs in two variables at once, so it bounds a region jointly and attributes nothing to either; and $\beta = 0.001$ was numerically inert ($\beta \cdot \text{KL} \leq 0.047$ reward units throughout), so this paper has no evidence about a KL penalty large enough to matter.

6 The ablation: the collapse is not the normalization default

We inherited `scale_rewards="group"`, verified as the trl 1.10.0 default. Reviewers of an earlier draft independently named it as a confound: near determinism, the within-group standard deviation dividing the advantage approaches zero, amplifying sampling noise into large updates — which alone might produce the collapse and its timing; Liu et al. [2025] argue against it on independent grounds.

We reran the 7B pipeline identically — same SFT warm start, learning rate, β, K, temperature, LoRA rank, epoch count, seed, and the same answer-ordered prompts of §2, so this ablation isolates the normalization, not the leak — with reward scaling disabled, the resolved value read back off the constructed trainer configuration. The signature is unchanged on the greedy $n = 108$ checkpoint sweep: validation semantic score rises to 62/432 at step 50, falls to 35/432 by step 100 and 4/432 by step 150, while structural validity climbs 0.70 to 0.96. The validation-selected peak (step 50, frozen before any test puzzle was scored) recovers 87/648 on test, $+0.377$ $[+0.247, +0.506]$ paired against base; the final policy 7/648, below base by -0.117 $[-0.179, -0.056]$ paired and within $+0.019$ of the original endpoint.

The signature survives in the dynamics too: without reward scaling, entropy falls to 0.0128 by step 150 (original endpoint 0.0099) and KL plateaus near 46 nats (original 47.05), intermediate magnitudes differing; the $n = 100$ temperature-0.9 sweep — a different measurement from the greedy checkpoint sweep above — reads 35/400 at step 50, 17/400 at 100, 2/400 from 150 on. Hump, collapse, entropy endpoint, below-base endpoint: all survive with the normalization off. The phenomenon is not a library default.

7 The control run: not memorization — a copy rule

Training-reward saturation says the policy solves its training prompts under sampling; was that SFT-stage memorization? The control: four arms, one vLLM session, greedy, both full splits, the same shuffled prompts as every other evaluation (Table 3).

Three results. The endpoint retains nothing — 21 of 3,228 training slots, indistinguishable from its own test floor: not a train/test gap, so not memorization. SFT does partially memorize, though less

Table 3: The control session: four arms, one vLLM session, greedy, both splits, shuffled evaluation prompts; percentile bootstrap, seed 0. Copy rate = emitted groups equal to consecutive quadruples of the prompt's word order (Appendix D).

Arm	Train groups (0–4)	of 3,228	Test groups (0–4)	of 648	Copy rate (test)
Instruct (untrained)	0.1871 [0.156, 0.224]	151	0.1605 [0.105, 0.222]	26	2.6%
SFT	0.5118 [0.461, 0.568]	413	0.3210 [0.216, 0.432]	52	0.3%
GRPO, step 50	0.6803 [0.616, 0.746]	549	0.4938 [0.383, 0.623]	80	0.2%
GRPO, step 403	**0.0260** [0.012, 0.041]	21	0.0247 [0.000, 0.056]	4	**92.4%**

Table 4: Above the rule: the leak-free session (four arms, one vLLM session, greedy, test split; conventions as Table 3; step 300 is the validation-selected peak, frozen before any test scoring). Below it: the sep05 session — seeds 1–2 of the repaired recipe and SFT trained on the leaked prompts — whose own base and SFT anchors reproduce 26 and 54.

Arm	Groups correct (0–4)	of 648	Invalid	Mean reward	Copy rate
Instruct (untrained)	0.1605 [0.105, 0.222]	26	6.8%	0.165	2.6%
SFT	0.3333 [0.228, 0.444]	54	22.2%	0.194	0.2%
GRPO-shuffled, step 300	0.8889 [0.741, 1.043]	144	6.2%	0.402	0.2%
GRPO-shuffled, step 403	**0.9136** [0.759, 1.074]	**148**	3.7%	**0.417**	0.2%
GRPO-shuffled, seed 1	0.8333 [0.691, 0.988]	135	5.6%	0.388	0.2%
GRPO-shuffled, seed 2	0.7963 [0.654, 0.951]	129	5.6%	0.370	0.2%
SFT on leaked prompts	0.0062 [0.000, 0.019]	1	0.0%	0.118	**100.0%**

than the raw numbers suggest: its $+0.191$ train-minus-test gap must be read against the untrained model's own $+0.027$ gap on the same chronological splits (split difficulty, not memorization); the adjusted excess is $+0.164$. Step 50's adjusted excess is $+0.160$ to SFT's $+0.164$ — a gap far inside the split-difference's uncertainty, so inheritance anywhere from none to all is consistent — while it improves on SFT by the same margin on both splits ($+0.169$ / $+0.173$): the peak's gain generalizes. And the shared session yields the paired stage split: $26 \rightarrow 52 \rightarrow 80 \rightarrow 4$, SFT $+0.1605$ $[+0.0556, +0.2716]$, RL $+0.1728$ $[+0.0741, +0.2840]$, endpoint -0.1358 $[-0.1975, -0.0802]$ vs. the untrained model (the original session read -0.136 $[-0.191, -0.080]$).

Four lines of string matching find the mechanism. A 1.6 training reward against 21 recovered training slots forced a look at the training prompts, where the bug of §2 was found. The behavioral test counts emitted groups equal, as sets, to consecutive quadruples of the prompt's word order (definition and full counts in Appendix D). Such quadruples are 91.4% of the groups the endpoint emits on training boards and 92.4% on test boards; 88.2% and 90.7% of its responses are pure four-quadruple copies; SFT and step 50 sit at 0.3%, below even the untrained model's weak bias (4.9% / 2.6%). The converged policy is a prompt-order copier. On training prompts, where order was the answer, copying earns 1.6; on shuffled prompts a copy is a fixed valid partition of a random order — exactly the uniformly drawn valid partition behind §3's 1.4-slot floor. Its pure-copy responses score no better than that process: 2 training slots against 6.25 chance-expected — a three-fold deficit, if anything, and certainly not an excess (we attach no test to it: a puzzle's four slots are not independent, Appendix B). What sits above chance is the aggregate — 21 slots against 7.1 expected; Table 3's puzzle-level interval, $[0.012, 0.041]$ on the 0–4 scale, excludes the chance mean of 0.0088, and test reads 4 against 1.4, same direction, interval not resolving — and it is carried almost entirely by responses that are *not* pure copies (35 non-copies recover 16 of the 21 slots; 60 partial copies, 3; 711 pure copies, 2): residual solving the rule has not displaced, not copying that works.

8 Reruns with the fix, and the leak without RL

We reran the reported recipe identically — same SFT warm start, hyperparameters, seed, trainer version, one epoch of 403 steps; only the logging cadence differs (Appendix E) — with the one-line fix applied, the constructed training set verified shuffled — by the same puzzle-seeded RNG as SFT and every evaluation — before any optimizer step. Everything this paper diagnoses disappears (Table 4).

The final checkpoint is the best policy in this study: 148 of 648 held-out slots — $5.7\times$ the untrained model — at $+0.7531$ $[+0.6111, +0.9074]$ paired against base and $+0.5802$ $[+0.4444, +0.7407]$ against SFT. And it is the *endpoint*: the validation curve climbs from 0.1065 at step 50 to 0.1944 at step 300 and plateaus (all nine checkpoints in Appendix E); final vs. validation peak is $+0.0247$ $[-0.0309, +0.0926]$ — there is no collapse to locate. Ten of 162 test puzzles are fully solved — the study's first meaningful solve rate — and the copy rule is gone (1 of 648 emitted groups, zero pure copies).

Every in-sample pathology vanishes with it. Over the 160 logged training rows the reward never touches 1.6 — never 1.0 (maximum 0.705, mean 0.371, rising by halves); `frac_reward_zero_std` peaks at 0.2 (mean 0.031) against the leaked run's sustained 1.0; the trainer's logged policy entropy never collapses (full range 0.116–0.212, a $1.8\times$ spread against the leaked run's 30-fold fall on the §5 sweep — different instruments, same verdict); and the trainer's logged KL stays below 0.23 throughout. Same reward function, same trainer: presented honestly, the task trains honestly, and the in-sample curve now sits near the held-out number (0.37 sampled vs. 0.417 greedy) instead of certifying a shortcut. The notebook's prespecified reading for this outcome, committed before launch: the leak was the cause, its in-sample signatures intrinsic to nothing about GRPO on this task. Two further seeds of the repaired recipe land at 135 and 129 of 648 ($+0.673$ $[+0.525, +0.815]$ and $+0.636$ $[+0.500, +0.772]$ paired against base), with no collapse anywhere: the 100–150 window, checkpointed every 10 steps on these seeds, climbs smoothly in both; validation peaks sit at steps 200 and 250 against seed 0's 300, and their test evaluations (127 and 132 of 648, in their own session) sit near the finals. Across three seeds the repaired endpoint reads 148, 135, 129 (the SFT arm decodes 52–56 across sessions; §3).

The leak, without RL. The remaining cell of the {leaked, clean} $\times$ {SFT, GRPO} design: SFT trained on the answer-ordered prompts — three epochs, the clean recipe otherwise — and evaluated on the same shuffled test split. It is a *perfect* copier: all 162 parsed responses are pure four-quadruple copies of their prompts (copy rate 1.000, past the leaked GRPO endpoint's 0.924), recovering 1 of 648 slots, -0.154 $[-0.222, -0.093]$ below the untrained model, at 0.0% invalid. Plain supervised learning suffices: the copying attractor belongs to the leaked task, not to the algorithm that trains on it. What GRPO specifically did was carry a *clean* initialization into that attractor (0.3% copy at its SFT init, 92.4% at its end) — and not all the way: the non-copy responses, the trace of its clean start, carry its residual slots (§7).

9 The reversal is legible from outside the reward

The leaked-run test generations from base, step 50 and the final policy were scored blind (board words and completion only) by three LLM judges (gpt-5-mini, gemini-2.5-flash, claude-sonnet-4-5), 1–10 scale, 1458/1458 parsed. All three rank step 50 above base above the final policy, all nine pairwise per-puzzle differences excluding zero under a paired bootstrap (seed 0; per-judge step-50-minus-final gaps $+0.60$ to $+3.25$). That ordering rules out the obvious confound: the final policy is the *most* well-formed arm on this split (0.6% invalid against 6.8% and 16.0%): a shared preference for well-formed text would have placed it first, not last — in hindsight, they were reading the copy rule (§7). Judges do err in correlated ways [Kohli, 2026], so we read this as three correlated readings of one signal — and of the reward's own semantic construct through another instrument: corroboration, not independent confirmation. The nine comparisons are one uncorrected family, all excluding zero by margins no correction would close.

10 Implications for evaluation practice

Distinguish in-sample from held-out proxy metrics before concluding a proxy has failed. Evaluated on held-out data, our reward ranks the three checkpoints correctly; it fails only in the form practitioners consult — the in-sample training curve, where it reads perfect. The cheap remedy is not a better reward but the same reward on data the policy was not updated against. Held out, the separation is real but not large ($+0.086$ paired at step 50, -0.040 at the endpoint, both excluding zero; SFT's does not) — and 79% of the endpoint's held-out reward is the cheap structural term, so a slightly different weighting could have kept the aggregate above base through the collapse; the decomposition below makes the catch a diagnostic rather than a coincidence.

Determinism certifies the scorer, not the task. Our reward was correct on every string it scored and the run was still gamed — through the prompt distribution; nothing in-sample prompted the four-line audit, held-out evaluation said there was something to find. One function, two measurements: in sample, reliable and invalid — this paper's title; held out, what validity it had.

Report reward components separately, never the aggregate. It is the one recommendation we would keep. The failure is invisible in the sum and obvious in the decomposition, which costs nothing: §5 reports it for our three arms (52%, 27%, 79%) — a standard reward-model diagnostic, simply not applied to deterministic rewards.

Assert prompt-path parity. The regression test that would have caught this bug asserts that every code path constructing a prompt yields the identical string per example (Appendix A); "not in answer order" would not have. The training distribution is code, so this parity check — not a dataset audit — is the shortcut-learning defense that transfers.

Treat single-seed RL results as uninterpretable. The caution applies to results that come out well too: the leaked run's step 50 is one seed, the argmax of a nine-point validation grid, selected before any test scoring (§3); the repaired recipe's three seeds are the caution heeded (§8).

11 Related work

Reward-model overoptimization. Gao et al. [2023] established this curve for *learned* reward models — gold-standard performance rising then falling with KL — as Goodhart against an approximation [Stiennon et al., 2020]; Karwowski et al. [2024] treat the curve formally for RL against proxies; Pan et al. [2022] report phase transitions shaped like the leaked 100–150 collapse (why that grid gap matters, §12). Our reward is a program: it approximates nothing, and the gaming ran through the prompt distribution (§2) — a channel a better reward model does not touch.

RL for reasoning and entropy collapse. Verifiable-reward RL underlies recent reasoning systems [DeepSeek-AI et al., 2025]; Yue et al. [2025] ask whether it extends capability beyond the base model's pass@k ceiling; our leaked run ends below both its antecedents, the repaired runs above them (§8). Shao et al. [2025] show Qwen2.5 gaining on MATH-500 under RLVR with random or incorrect rewards; part of the leaked step-50 gain may be that phenomenon. Cui et al. [2025] document entropy collapse as a central dynamic of RL for reasoning; here it coincides with convergence onto a leaked shortcut — symptom, not cause: leak removed, entropy never collapses (§8). Liu et al. [2025] critique group-level normalization; §6 shows independence from it.

Reward hacking and its taxonomy. Skalse et al. [2022] formalize the failure mode (as *reward gaming*); ours is gaming of the composed task — reward plus prompt distribution — not of the reward function, which was exact. Baker et al. [2025] show the failure at frontier scale: reasoning models hack verifiable rewards and obfuscate it under monitor pressure. Krakovna et al. [2020] catalogue specification gaming across RL systems; our entry is ordinary — the trace is the novelty.

Shortcut learning and leakage. What the leaked policy learned is a shortcut [Geirhos et al., 2020]: a rule that succeeds in training because an input feature — word order — carried the label; Lapuschkin et al. [2019]'s Clever Hans, in vision. Neither usually reaches RL post-training, where the leak hides in code (§10) — our transfer.

Validity in AI evaluation. Measurement-theoretic critiques [Jacobs and Wallach, 2021, Freiesleben and Zezulka, 2025, Chehbouni et al., 2025, Manheim and Garrabrant, 2018] distinguish a measurement from its construct; ours is maximally reliable and the validity gap does the damage anyway.

12 Limitations

The leaked-SFT cell (§8) is one seed; every run shares one task and reward; no human baseline exists (only "worse than untrained" is established). An endpoint cannot rule out a traversed 1.5B peak (§4); the leaked peak may move across seeds (§3); a meaningful KL penalty is untested; the 1.5B loader shares the leak, copy rate unmeasured; the judges are correlated scorers (§9).

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A The two prompts

The bug of §2 is one line, and it is easiest to see as the pair of strings the model actually received. Both are the user turn of the same puzzle (id 3, 2023-06-14), under the same system prompt; the answer key is `{CHEEK, EYE, MOUTH, NOSE}`, `{CHOW, GOBBLE, SCARF, WOLF}`, `{LAB, PEKE, PIT, POM}`, `{AMIGO, KING, STOOGE, TENOR}`.

GRPO training prompt, as the reported run consumed it (`train/grpo.py::build_dataset`):

```
Words: CHEEK, EYE, MOUTH, NOSE, CHOW, GOBBLE, SCARF, WOLF,
       LAB, PEKE, PIT, POM, AMIGO, KING, STOOGE, TENOR
```

SFT data and every evaluation (`data/formatting.py::shuffled_words`, a puzzle-seeded permutation):

```
Words: CHOW, POM, PIT, KING, AMIGO, NOSE, SCARF, CHEEK,
       EYE, TENOR, STOOGE, GOBBLE, MOUTH, LAB, PEKE, WOLF
```

In the first, emitting the words four at a time in the order given scores the full 1.6. In the second it scores what a uniformly drawn valid partition scores. Nothing else differs — same system prompt, same reward, same parser — and the difference is a missing call to the shuffle the rest of the codebase already used. The regression test we have since added asserts the two paths agree per puzzle, which is the property that would have caught this and the weaker "is not in answer order" assertion would not.

B Per-checkpoint values for the entropy sweep

Table 5 tabulates all eleven rows of the sweep artifact: the trajectory plotted in Figure 1 (including the step-400 checkpoint, indistinguishable from step 403 at this precision), the structural column discussed in Section 5, and the off-trajectory untrained Instruct reference. All values are from the $n = 100$ temperature-0.9 entropy sweep of the reported 7B run (seed

0); the artifact is `results-analysis/entropy-kl-7b.json` in the released repository, `https://github.com/jacksonmlukas/connections-rl`, which also holds the training and evaluation configs, the reward implementation, and the generation logs behind every table here.

Unless stated otherwise, all intervals in this paper are percentile bootstrap over *puzzles* — the unit of independent sampling — with $B = 1{,}000$ resamples at seed 0 (practice motivated by Agarwal et al., 2021); per-slot counts are reported alongside means but never resampled independently, since a puzzle's four slots are not independent. Where a count is small enough that the bootstrap degenerates (Table 6), we substitute Clopper–Pearson intervals and say so.

Table 5: The reported 7B run on the $n = 100$ temperature-0.9 entropy sweep. Structural = fraction of outputs that are a valid partition; semantic = mean correct groups / 4. KL is the per-sequence sample-based estimate against the SFT initialization, each row estimated under its own policy's samples; the untrained Instruct row is off the GRPO trajectory (§5).

Checkpoint	Policy entropy	KL from SFT init	Structural	Semantic
Instruct (untrained)	0.2577	42.60	0.69	0.0125
SFT init	0.3025	0.00	0.37	0.0325
Step 50	0.2107	2.70	0.68	**0.0950**
Step 100	0.1999	8.12	0.64	0.0875
Step 150	0.0158	46.42	0.95	0.0075
Step 200	0.0109	46.88	0.95	0.0100
Step 250	0.0102	46.96	0.96	0.0100
Step 300	0.0099	47.05	0.96	0.0100
Step 350	0.0099	47.05	0.96	0.0100
Step 400	0.0099	47.05	0.96	0.0100
Step 403	0.0099	47.05	0.96	0.0100

C 7B endpoints across three seeds

Table 6 tabulates the three-seed endpoint session discussed in Section 3; the 4 / 7 / 11 spread is the basis of the single-seed caveats in Sections 10 and 12. The pass@16 contrasts referenced there are differences of $+0.920$ $[0.778, 1.062]$ for SFT over GRPO and -0.370 $[-0.475, -0.259]$ for GRPO against the untrained Instruct model — single-seed, read as illustrative only.

Table 6: 7B endpoints across three seeds (separate serving session from Table 2; the SFT arm decodes to 0.321/52 both here and in the Table 2 control session, and to 0.346/56 in a third session; the untrained and final-checkpoint anchor arms reproduce across sessions exactly).

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward
Instruct (untrained)	0.160	26	6.8% $[3.1, 11.1]$	0.165
SFT	0.321	52	22.2%	0.197
GRPO (3 seeds)	0.045 ± 0.022	4 / 7 / 11	$1.2 \pm 0.6\%$	0.132 ± 0.008

D Control-session copy rates and rewards

Table 7 gives the full copy-rule test behind Section 7: for each generation, an emitted group counts as a copy when it equals, as a set, a consecutive quadruple (positions 1–4, 5–8, 9–12, 13–16) of that prompt's word order. Train prompts parsed: 806 of 807 per arm; test: 162 of 162. Mean held-out rewards in this session: base 0.1802 (train) / 0.1654 (test), SFT 0.2490 / 0.1910, step 50 0.3113 / 0.2519, endpoint 0.1242 / 0.1250 — the endpoint's test values reproduce the published 0.125 exactly. The per-stratum relative loss cited in §3 (untrained model $\rightarrow$ endpoint, over the 158 of 162 test puzzles with a single stratum label): 92% on `category` ($n = 69$), 67% on `cultural` ($n = 28$), 86% on `tag-fillin` ($n = 40$), 100% on `wordplay` ($n = 21$, from a near-floor base of 0.048, one recovered slot).

Table 7: Copy-rule rates, control session (greedy; shuffled evaluation prompts). Group-level = share of emitted groups that are prompt-order quadruples; response-level = share of responses whose four groups all are.

Arm	Train (group)	Train (response)	Test (group)	Test (response)
Instruct (untrained)	4.9% (159/3224)	4.3%	2.6%	1.2%
SFT	0.3% (9/3224)	0.0%	0.3%	0.0%
GRPO, step 50	0.3% (11/3224)	0.0%	0.2%	0.0%
GRPO, step 403	**91.4%** (2946/3224)	**88.2%**	**92.4%**	**90.7%**

E Leak-free rerun: checkpoint curve and training signals

Validation checkpoint curve of the §8 rerun ($n = 108$ validation puzzles, greedy; artifact `results-analysis/aug29/ckpt-curve-7b-shuffled.json`, evaluation outputs under `results-analysis/aug29/leakfree-session-test/`, and the full training log alongside them):

Step	Structural	Semantic (0–1)	Solve rate	Mean reward
50	0.852	0.1065	0.000	0.229
100	0.954	0.1574	0.028	0.331
150	0.972	0.1782	0.019	0.356
200	0.963	0.1736	0.009	0.351
250	0.954	0.1806	0.028	0.368
300	0.944	**0.1944**	0.028	0.376
350	0.963	0.1921	0.028	0.377
400	0.972	0.1921	0.028	0.381
403	0.972	0.1921	0.028	0.380

Training-signal summary over the 160 logged rows (single T4, QLoRA, 1.8×10^4 s for 403 steps): reward min/mean/max 0.123/0.371/0.705, rising 0.30 $\rightarrow$ 0.44 by halves; `frac_reward_zero_std` max 0.2, mean 0.031; across-rollout reward std is never zero on any logged row; trainer-logged policy entropy 0.116–0.212 and KL at most 0.22 (trainer instruments, not directly comparable to the §5 sweep estimates). One nonsubstantive difference from the leaked run: logging cadence (5-step averages there, 160 rows over 403 steps here), so cross-run in-sample extrema are extrema over differently smoothed series.

Seeds 1–2 (artifacts under `results-analysis/sep05/`): validation semantic score at steps 50/100/110/120/130/140/150/200/250/300/350/400/403 reads 0.141/0.160/0.162/0.157/0.171/0.190/0.194/**0.227**/0.215/0.220/0.218/0.220/0.220 for seed 1 and 0.130/0.162/0.169/0.174/0.185/0.185/0.197/0.199/**0.206**/0.183/0.185/0.190/0.194 for seed 2 — smooth climbs through the densely checkpointed 100–150 window, validation peaks at steps 200 and 250, plateaus thereafter. Test evaluations of those peaks read 127/648 and 132/648 against base's 26 in their own session — close to their finals, though the two sit in different serving sessions, so no paired contrast is available. The leaked-SFT arm's evaluation and the seed finals share the sep05 session with base and SFT anchors at 26 and 54 of 648.

F 1.5B test-split table

Table 8 tabulates the 1.5B arms discussed in Section 4; every mean is the exact ratio of its count to 162.

Table 8: 1.5B on the test split (162 puzzles, 648 group slots). Groups-correct intervals are Clopper–Pearson exact binomial on the slot count of 648, scaled to 0–4 (a percentile bootstrap is degenerate at counts of 1–2); slots cluster within puzzles, so they are approximate. Invalid-rate intervals are percentile bootstrap over puzzles. Conditioned on structurally valid output (§3), the arms read: base 0.009 ($n = 110$ of 162), SFT 0.048 ($n = 42$), GRPO 0.006 ($n = 158$).

Arm	Groups correct (0–4)	Groups (of 648)	Invalid	Mean reward	Solves
Base	0.0062 $[0.0002, 0.0343]$	1	32.1% $[24.7, 38.9]$	0.049	0.0%
SFT	0.0123 $[0.0015, 0.0444]$	2	74.1% $[67.3, 80.2]$	-0.038	0.0%
GRPO	0.0062 $[0.0002, 0.0343]$	1	2.5% $[0.6, 5.6]$	0.113	0.0%